LESSON 6.5

Write and Evaluate Algebraic Expressions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an algebraic expression for a real situation and evaluate it for a given value of the variable.

WHY THIS MATTERS Evelyn is helping to plan the sixth-grade field day. The committee will order the items shown for each student participating in field day. They do not yet know how many students will be participating. The shipping cost for all items is a one-time cost of $23.50. What expression can Evelyn use to determine how much the committee will need to purchase the items?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Variable *Spanish: Variable*	A letter that stands for a number that is unknown or can change.	the m in 14m + 35
Algebraic Expression *Spanish: Expresión algebraica*	A math phrase that has at least one letter.	14m + 35
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	In 14m, the coefficient is 14.
Constant *Spanish: Constante*	A fixed number in an expression or equation that does not change.	the 35 in 14m + 35
Like terms *Spanish: Términos semejantes*	Terms with the same letter raised to the same power, like 2x and 5x.	4x + 2x = 6x (like terms, same variable); but 4x + 2y cannot be combined (different variables)
Substitute *Spanish: Sustituir*	To replace a variable with a given number before working out the value.	For s = 30: 4.25s + 23.50 → 4.25 × 30 + 23.50

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A taxi charges $3 to start plus $2 for each mile. Write an expression for the fare for m miles.

1. Let m = the number of miles.
2. $2 repeats each mile: 2m.
3. The $3 start fee happens once. Join: 2m + 3.

Answer: 2m + 3 dollars

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The field day committee orders one $4.25 item for each student, plus a one-time $23.50 shipping fee. The number of students is unknown.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I let _____ stand for _____ .
- The coefficient _____ shows the amount for each _____ .
- The constant _____ happens only _____ .
- My expression for this situation is _____ .

Word bank variable | algebraic expression | coefficient | constant | term | each | one-time | per | stands for | define | total | unknown

WATCH OUT!

A common mistake

A common mistake in Write Algebraic Expressions is treating 'each' or 'per' as if it means add instead of multiply, and attaching the variable to the wrong number. For 'a $25 setup fee plus $8 per hour, h,' students often write 25 + 8 + h (adding all three numbers) or 25h + 8 (multiplying the flat fee by h) instead of the correct 8h + 25. Since $8 applies per hour, it must multiply the variable (8h), and the flat $25 fee is a constant added only once

LESSON 6.5

Write and Evaluate Algebraic Expressions

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression represents 'the product of 6 and a number n'?

(A) 6n
(B) 6 + n
(C) n − 6
(D) n ÷ 6

HINT Underline the key word 'product.' Every operation has a code word — which operation does 'product' name?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which expression represents '9 less than a number y'?

(A) y − 9
(B) 9 − y
(C) y + 9
(D) 9y

HINT 'Less than' is a tricky phrase — it flips the order of what you write.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which expression represents 'the sum of a number m and 15'?

(A) m + 15

(B) m − 15

(C) 15m

(D) m ÷ 15

HINT Circle the key word 'sum.' Which operation gives you a sum?

SHOW YOUR WORK

4 **Write the name of the correct group on each line.**

Sort each key word by the operation it represents.

Groups: **Addition** **Subtraction** **Multiplication**

_____ sum

_____ more than

_____ difference

_____ less than

_____ product

_____ times

HINT Each word is a code for add, subtract, or multiply. Think about what result the word describes.

PART 2 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

Evaluate 4.25s + 23.50 when s = 10.

(A) $47.75

(B) $66.00

(C) $66.75

(D) $277.50

HINT Replace s with 10.

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

6 **Circle the letter of the best answer. Show how you know.**

Which key word tells you to multiply?

(A) Product

(B) Sum

(C) Difference

(D) Quotient

HINT Think of a fact like 3 × 4 = 12. What do we call the number 12 in that fact?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What key words helped you decide which operation to use? How can the same phrase be written differently?

SENTENCE STARTER The phrase '___ ' tells me to use ___ because the key word ___ means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

LESSON 6.5

Write and Evaluate Algebraic Expressions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Match each word phrase to its correct algebraic expression.

Groups: Correctly Matched

_____ twice a number k → 2k

_____ a number b divided by 5 → b ÷ 5

_____ the sum of a number d and 15 → d + 15

_____ 10 decreased by a number r → 10 − r

_____ 4 more than triple a number c → 3c + 4

_____ half of a number h → h ÷ 2

2 **Write the letter of the matching item on each line.**

Match each word phrase to its algebraic expression.

_____ the product of 5 and n

_____ 8 less than a number w

_____ a number p increased by 12

_____ twice a number x, plus 3

_____ a number y divided by 4

_____ 7 more than a number z

(A) 5n

(B) w − 8

(C) p + 12

(D) 2x + 3

(E) y ÷ 4

(F) z + 7

3 **Complete the table. Show how you found each value.**

Write an algebraic expression for each real-world situation.

Situation	Variable	Expression
Movie tickets cost $12 each	t = number of tickets	
You had $50 and spent some money	s = dollars spent	
Each student gets 3 pencils plus 2 erasers	n = number of students	
A pizza is split equally among friends	f = number of friends, pizza = 24 slices	

SHOW YOUR WORK

4 **Decide whether each pair balances. Check Yes or No, then explain.**

Both expressions are equivalent by the commutative property of addition.

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

The expression 8n + 15 gives the cost of n tickets plus a fee. What is the cost for 6 tickets?

(A) $29
(B) $48
(C) $63
(D) $138

My work — one step per line	What I did to both sides

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Phrase** 8 divided by a number m
2. **Student's expression** m ÷ 8
3. **Reasoning** I saw 'divided by' so I wrote the variable first and then 8.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What key words helped you decide which operation to use? How can the same phrase be written differently?

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LESSON 6.5

Write and Evaluate Algebraic Expressions

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Translate each two-step word phrase into an algebraic expression. Be careful with the order!

Word Phrase	Algebraic Expression
5 more than triple a number x	
twice the sum of a number y and 4	
10 less than the product of 7 and a number z	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Translation Error

1. **Phrase** 12 less than a number g
2. **Student's expression** 12 − g
3. **Reasoning** I see 'less than' so I subtract, and 12 comes first in the phrase.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Field day needs one $4.25 item per student plus a one-time $23.50 shipping fee. Write the expression, then find the cost for 30 students AND for 60 students. Does doubling the students double the cost? Explain.

WRITE YOUR ANSWER

4 Answer in complete sentences.

Write a real-world situation that could be modeled by the expression 4n + 10. Explain what the variable n represents, what the coefficient 4 means, and what the constant 10 represents.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What key words helped you decide which operation to use? How can the same phrase be written differently?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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